ICAD RT-5 · Set 7 — Answer Sheet

Exam 16-11-2026 · 75 Q · 300 marks · 3 hours · +4 correct, −1 wrong, 0 blank

Name ..... Roll No ..... Date .....

Physics

1	a	b	c	d
2	a	b	c	d
3	a	b	c	d
4	a	b	c	d
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18	a	b	c	d
19	a	b	c	d
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Numerical — write the value

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Chemistry

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20	a	b	c	d

Numerical — write the value

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Mathematics

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18	a	b	c	d
19	a	b	c	d
20	a	b	c	d

Numerical — write the value

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Bubbles — single correct · Box — write the value · each subject out of 100

ICAD RT-5 · Set 7

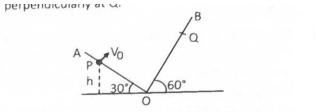
Exam 16-11-2026 · 25 Physics · 25 Chemistry · 25 Mathematics · 300 marks · 3 hours · +4 / −1

Physics

PART A · Q1–20 · SINGLE CORRECT

1. P4L3V5Q3 A particle is released from rest from a tower of height 3h. The ratio of times to fall equal heights h, i.e. t_1:t_2:t_3, is: (a) √3:√2:1 (b) 3:2:1 (c) 9:4:1 (d) 1:(√2−1):(√3−√2)

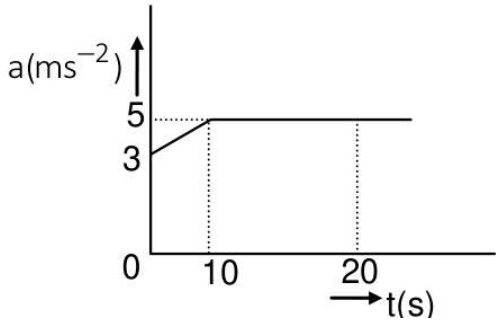
2. P4L7V5Q9 Two inclined planes OA and OB are shown in the figure (OA makes 30° and OB makes 60° with the horizontal through O, on opposite sides). A small ball is thrown from the point P on OA (at height h above O's level) with speed $v_0 = 10\sqrt{3}$ ms^{-1} along a direction perpendicular to plane OA and it hits the inclined plane OB perpendicularly at Q (take g = 10 ms^{-2}). Match the columns: Column-I: (i) Time of flight of the ball (in second) (ii) Speed (in ms^{-1}) with which the ball hits the plane OB (iii) The value of h (in metre) (iv) The maximum height (in metre) from O attained by the ball. Column-II: (p) 2 (q) 5 (r) 16.25 (s) 10. (a) (i)–p, (ii)–q, (iii)–r, (iv)–s (b) (i)–q, (ii)–r, (iii)–p, (iv)–s (c) (i)–p, (ii)–s, (iii)–q, (iv)–r (d) (i)–pr, (ii)–qs, (iii)–qs, (iv)–rs



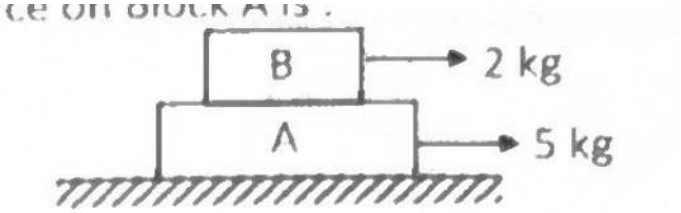
Match the following columns and mark the correct option from the codes given below

	Column-I		Column-II
(i)	Time of flight of the ball (in second)	(p)	2
(ii)	Speed (in ms^{-1}) with which the ball hits the plane OB	(q)	5
(iii)	The value of h (in metre)	(r)	16.25
(iv)	The maximum height (in metre) from O attained by the ball	(s)	10

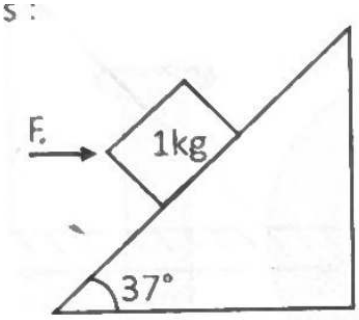
3. P4XVRT2Q14 The graph describes an airplane's acceleration during its take-off run (starting from rest). The airplane's velocity when it lifts off at t = 20 s is: (a) 40 m s^{-1} (b) 50 m s^{-1} (c) 90 m s^{-1} (d) 180 m s^{-1}



4. P5L2V3Q3 Block B (2 kg) rests on block A (5 kg), which rests on the ground (see figure). The net force on Block A is: (a) 20 N (b) 70 N (c) zero (d) 50 N

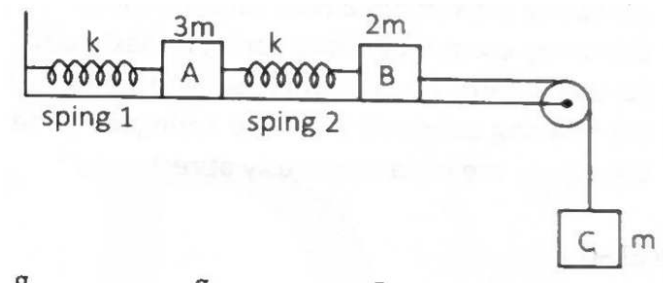


5. P5L3V6Q6 A 1 kg block rests on a smooth inclined surface of inclination 37°, held at rest by a horizontal force F (see figure; g = 10 m/s², sin37° = 3/5). The normal force between the block and the inclined surface is: (a) 15 N (b) 10 N (c) 7.5 N (d) 12.5 N

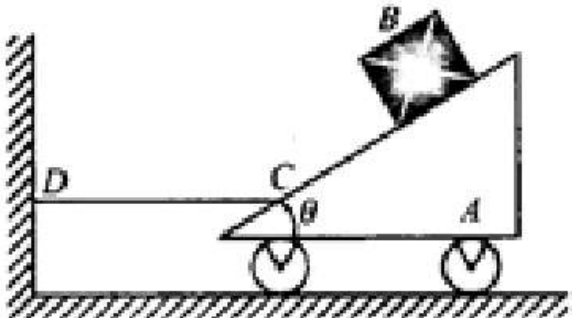


6. P6L2V1Q5 "When a hanging carpet is beaten with a stick, dust particles start coming out of it." This phenomenon is best example of: (a) Newton's First Law of motion (b) Newton's Second Law of motion (c) Newton's Third Law of motion (d) Newton's Law of gravitation

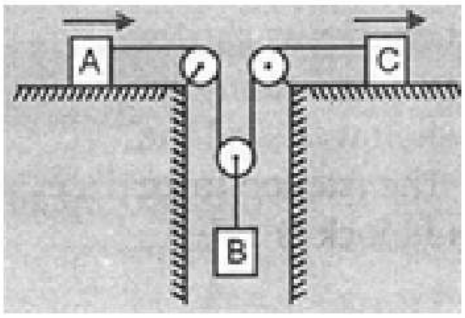
7. P6L6V5Q6 Three blocks are connected as shown in the figure on a smooth horizontal table: wall — spring 1 (k) — block A (3m) — spring 2 (k) — block B (2m) — light string over a smooth pulley at the table edge — block C (m) hanging. The system is in equilibrium. Find the acceleration of block A if spring (2) is cut suddenly. (a) g/3 (b) g/2 (c) g/6 (d) none



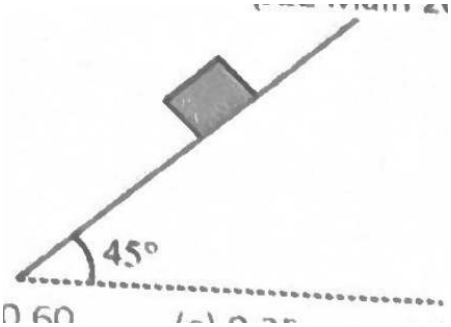
8. P6XVRT3Q13 Block B has mass 'm' and is released from rest when it is on top of wedge A, which has a mass 3m. Determine the tension in cord CD needed to hold the wedge from moving while B is sliding down A. Neglect friction. (a) 2mg cosθ (b) (mg/2) cosθ (c) (mg/2) sin 2θ (d) (mg/2) sinθ



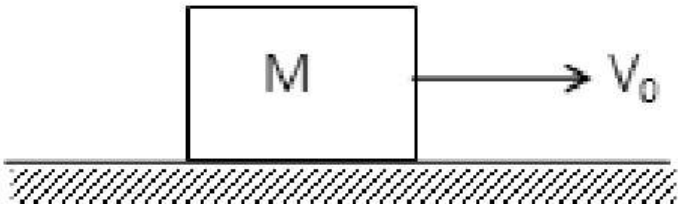
9. P6XVRT4Q20 Blocks A and C start from rest and move to the right with accelerations a_A = 12t m/s² and a_C = 3 m/s². Here 't' is in seconds. The time when block B again comes to rest is : (a) 1 s (b) 3/2 s (c) 2 s (d) 1/2 s



10. P7SBVPYQQ10 Consider a block kept on an inclined plane (inclined at 45°). If the force required to just push it up the incline is 2 times the force required to just prevent it from sliding down, the coefficient of friction between the block and inclined plane (μ) is equal to: (a) 0.33 (b) 0.60 (c) 0.25 (d) 0.50



11. P7XVCAT5Q12 Find the distance travelled by the block if μ_K = Vot/2 and initial velocity of block is V0? (a) infinite (b) 4V0/(3√g) (c) V0²/g (d) 2V0/g



12. P8L1V3Q4 A person holds a bucket of weight 60 N. He walks 7 m along the horizontal path and then climbs up a vertical distance of 5 m. The work done by the man (on the bucket) is: (a) 300 N-m (b) 420 N-m (c) 720 N-m (d) none of these

13. P8L2V4Q2 Force acting on a particle is (2î + 3ĵ) N. Work done by this force is zero, when a particle is moved on the line 3y + kx = 5. Here value of k is (a) 2 (b) 4 (c) 6 (d) 8

PART B · Q21–25 · NUMERICAL VALUE

21. P3XVRT1Q22 For three vectors A = (−xî − 6ĵ − 2k̂), B = (−î + 4ĵ + 3k̂) and C = (−8î − ĵ + 3k̂), if A·(B×C) = 0, then x is ____.

14. P9L4V6Q1 A particle of mass m is moving with speed 2v and collides with a mass 2m moving with speed v in the same direction. After collision, the first mass is stopped completely while the second one splits into two particles each of mass m, which move at angle 45° with respect to the original direction. The speed of each of the moving particle will be (a) √2v (b) v/√2 (c) v/(2√2) (d) 2√2v

15. P10L1V5Q6 Passage (Q6 to Q8): A small sphere of mass m suspended by a thread is first taken aside, so that the thread forms the right angle with the vertical and then released. — The total acceleration of the sphere and the thread tension as function of θ, the angle of deflection of the thread from the vertical will be (a) g√(1 + 3cos²θ), T = 3mg cosθ (b) g cosθ, T = 3mg cosθ (c) g√(1 + 3sin²θ), T = 5mg cosθ (d) g sinθ, T = 5mg cosθ

16. P10L3VSuggQ1 What is the direction of centripetal acceleration in circular motion? (a) Away from the center (b) Tangential to the circle (c) Perpendicular to the radius (d) Towards the center along the radius

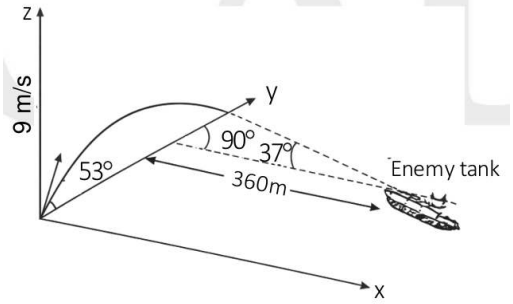
17. P10L6L7V4Q10 Multiple Correct Answer Question. For a curved track of radius R, banked at angle θ (Take v 0 = √(Rg tanθ)): (one or more options may be correct) (a) a vehicle moving with a speed v 0 is able to negotiate the curve without calling friction into play at all (b) a vehicle moving with any speed V > V 0 is always able to negotiate the curve, with friction called into play (c) a vehicle moving with any speed V < V 0 must have the force of friction into play (d) the minimum value of the angle of banking for a vehicle parked on the banked road can stay there without slipping, is given by θ = tan −1 μ 0 (μ 0 = coefficient of static friction)

18. P10SBVPYQQ4 A car of mass m moves on a banked road having radius r and banking angle θ. To avoid slipping from the banked road, the maximum permissible speed of the car is v 0 . The coefficient of friction μ between the wheels of the car and the banked road is: (a) μ = (v 0 ²+rg tanθ)/(rg+v 0 ²tanθ) (b) μ = (v 0 ²−rg tanθ)/(rg−v 0 ²tanθ) (c) μ = (v 0 ²−rg tanθ)/(rg+v 0 ²tanθ) (d) μ = (v 0 ²+rg tanθ)/(rg−v 0 ²tanθ)

19. P11L2V1Q1 Four point masses P, Q, R and S with respective masses 1 kg, 1 kg, 2 kg and 2 kg form the corners of a square of side a, taken in order P, Q, R, S around the square. The centre of mass of the system will be farthest from (a) P only (b) R and S (c) R only (d) P and Q

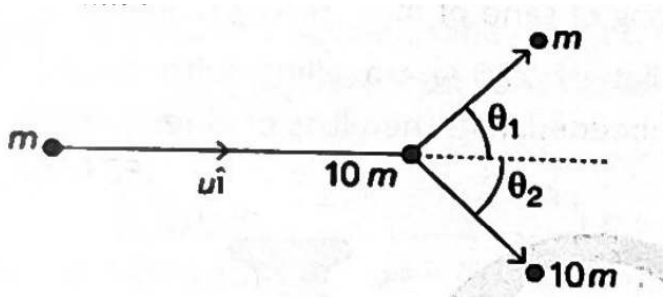
20. P11L3V2Q1 A uniform square lamina of side 6 m has its centre at the origin. A small square of side 2 m is removed from the top-right corner. Find the x-coordinate of the new centre of mass of the remaining lamina. (a) −(1/4) m (b) +(1/4) m (c) −(1/3) m (d) 0

22. P4XVRT2Q25 Figure shows an enemy tank moving with a speed of 9 m/s in the direction shown in figure (at 37° to the 360 m perpendicular, towards the plane of firing). A gun at the origin can fire shells in the y-z plane only, with a speed 100 m/s at an angle of 53° with the horizontal. Tank is initially at a perpendicular distance of 360 m from the plane of firing, and its path meets that plane at the point where the shell lands. If the tank started at t = 0, then after what time interval (in s) is the shell to be fired to hit the tank? (g = 10 m s^{-2})



23. P8L3V3Q3 If the momentum of a body is increased by 50%, the percentage increase in its kinetic energy is (keeping mass constant)

24. P9L5L6V6Q3 A particle of mass m is moving along the X-axis with initial velocity ui. It collides elastically with a particle of mass 10m at rest and then moves with half its initial kinetic energy (see figure). If $\sin\theta_1=\sqrt{n}\sin\theta_2$, then value of n is _____.



25. P11SBVPYQQ5 The position vectors of two 1 kg particles, (A) and (B), are given by $\bar{r}\,A = (\alpha\,1\,t\,2\,\hat{i} + \alpha\,2\,t\hat{j} + \alpha\,3\,t\,\hat{k})$ m and $\bar{r}\,B = (\beta\,1\,t\hat{i} + \beta\,2\,t\,2\,\hat{j} + \beta\,3\,t\,\hat{k})$ m, respectively; ($\alpha\,1 = 1$ m/s 2 , $\alpha\,2 = 3n$ m/s, $\alpha\,3 = 2$ m/s, $\beta\,1 = 2$ m/s, $\beta\,2 = -1$ m/s 2 , $\beta\,3 = 4p$ m/s), where t is time, n and p are constants. At t = 1 s, $|\bar{v}\,A| = |\bar{v}\,B|$ and velocities $\bar{v}\,A$ and $\bar{v}\,B$ of the particles are orthogonal to each other. At t = 1 s, the magnitude of angular momentum of particle (A) with respect to the position of particle (B) is $\sqrt{L}$ kg m 2 s −1 . The value of L is _____.

Chemistry

PART A · Q1–20 · SINGLE CORRECT

1. C1L9V4Q8 a and b are van der Waals constants for gases. Chlorine is more easily liquefied than ethane because — (a) a and b for Cl_2 > a and b for C_2H_6 (b) a and b for Cl_2 < a and b for C_2H_6 (c) a for Cl_2 < a for C_2H_6 but b for Cl_2 > b for C_2H_6 (d) a for Cl_2 > a for C_2H_6 but b for Cl_2 < b for C_2H_6

2. C2L8VSuggQ3 A neutron has kinetic energy K = 3.2×10^{-21} J. Calculate its de-Broglie wavelength. Use $m_n = 1.67\times10^{-27}$ kg (h = 6.626×10^{-34} J s). (a) 1.05×10^{-10} m (b) 2.02×10^{-10} m (c) 3.27×10^{-12} m (d) 6.40×10^{-11} m

3. C2L10VSuggQ2 Which set of quantum numbers represents an electron in a 4p orbital? (a) n = 4, l = 1 (b) n = 4, l = 2 (c) n = 3, l = 1 (d) n = 2, l = 1

4. C3L2VSuggQ2 How many mole of atoms are in a mole of CH_3COOH molecule? (a) 2 moles of C atoms, 4 moles of H atoms, 2 moles of O atoms (b) 1 mole of C atom, 2 moles of H atoms, 1 mole of O atom (c) 2 moles of C atoms, 3 moles of H atoms, 2 moles of O atoms (d) None of these

5. C3L8V5Q7 Calculate the molarity of water if its density is 1000 kg/m^3 (a) 53.55 M (b) 55.55 M (c) 54.50 M (d) 50.55 M

6. C4L1V6Q4 In which group would you place an element with Z = 117? (a) Group 13 (b) Group 15 (c) Group 17 (d) Group 18

7. C4L3V5Q8 The size of the species Pb, Pb^{2+}, Pb^{4+} decreases as: (a) $Pb^{4+} > Pb^{2+} > Pb$ (b) $Pb > Pb^{2+} > Pb^{4+}$ (c) $Pb > Pb^{4+} > Pb^{2+}$ (d) $Pb^{4+} > Pb > Pb^{2+}$

8. C5L3V3Q2 What is the hybridisation and bond angle of $BeCl_2$? (a) sp, 180° (b) sp^2, 120° (c) sp^3, 109.5° (d) dsp^2, 90°

9. C5L3V5Q7 A covalent bond is formed between the atoms by the overlapping of orbitals containing: (a) single electron (b) paired electron (c) single electron with parallel spin (d) single electron with opposite spin

10. C5L5V3Q5 In PF_5, find hybridisation and geometry (5 bond pairs, 0 lone pairs). (a) sp^3d, Trigonal bipyramidal (b) sp^3, Tetrahedral (c) sp^3d^2, Octahedral (d) sp^2, Trigonal planar

11. C5L6V6Q3 The correct order of bond angles is: (a) $H_2S < NH_3 < BF_3 < SiH_4$ (b) $NH_3 < H_2S < SiH_4 < BF_3$ (c) $H_2S < NH_3 < SiH_4 < BF_3$ (d) $H_2S < SiH_4 < NH_3 < BF_3$

12. C6L1V4Q7 An isolated system is that system in which: (a) there is no exchange of energy with the surrounding (b) there is no exchange of mass with the surroundings (c) there is no exchange of energy and mass with the surroundings (d) there is exchange of mass with the surroundings

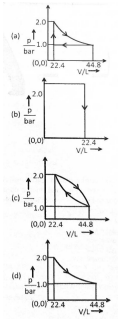
13. C6L2V5Q6 Which statements correctly describe characteristics of an irreversible process? 1. It takes place in a finite number of steps. 2. Equilibrium exists only at initial and final stages. 3. Work obtained is maximum. 4. It cannot be reversed without causing permanent change in surroundings. (a) 1, 2, and 4 only (b) 1 and 3 only (c) 2 and 3 only (d) All four statements

14. C6L3V3Q4 A gas is expanded isothermally and reversibly from volume 1 L to 10 L at 298 K. Calculate work done using $W = 2.303nRT\log(V_f/V_i)$, n=1 mol. (a) 5745 J (b) 5706 J (c) 5900 J (d) 5400 J

15. C6L3V5Q2 When freezing of a liquid takes place in a system: (a) may have q > 0 or q < 0 depending on the liquid (b) is represented by q > 0 (c) is represented by q < 0 (d) has q = 0

16. C6L4V4Q1 Which reaction is endothermic in nature? (a) $CaCO_3 \rightarrow CaO + CO_2$ (b) $C + O_2 \rightarrow CO_2$ (c) $NaOH + HCl \rightarrow NaCl + H_2O$ (d) $C(graphite) + O_2(g) \rightarrow CO_2(g)$

17. C6SBVPYQQ4 Which of the following graphs between pressure 'P' versus volume 'V' represent the maximum work done? (a) a cycle: reversible isothermal curve from (22.4 L, 2.0 bar) to (44.8 L, 1.0 bar), then back at constant 1.0 bar to 22.4 L, then up at constant volume to 2.0 bar (b) expansion at a constant 2.0 bar from V = 0 to 22.4 L, then a constant-volume drop at 22.4 L (c) a cycle between two curves joining (22.4 L, 2.0 bar) and (44.8 L, 1.0 bar) (d) a single reversible isothermal expansion curve from (22.4 L, 2.0 bar) to (44.8 L, 1.0 bar) [see figure]



PART B · Q21–25 · NUMERICAL VALUE

21. C1XVRT2Q32 Research is often carried out at very low pressure in an ultra-high-vacuum chamber (UHV), where a gas behaves ideally. At a pressure p = 3 × 10 −15 bar, a volume V = 1.0 cm³ and 300 K, how many molecules are there? (R = 0.0821 L atm K −1 mol −1 , N A = 6.022 × 10 23) (A) 600 (B) 60,000 (C) 6 × 10 6 (D) 6 × 10 9

22. C2XVCAT3Q48 Calculate the principal quantum number n for a Li^{2+} ion if it is known that 15 different types of photons can be emitted when the ion transitions from an excited state to lower energy levels.

18. C6SBVPYQQ5 Consider the following data: Δ f H°(methane, g) = −X kJ mol −1 ; enthalpy of sublimation of graphite = Y kJ mol −1 ; dissociation enthalpy of H 2 = Z kJ mol −1 . The bond enthalpy of the C–H bond is given by: (a) (X + Y + 2Z)/4 (b) (X + Y + 4Z)/2 (c) X + Y + Z (d) (−X + Y + Z)/4

19. C6SBVPYQQ9 20.0 dm 3 of an ideal gas 'X' at 600 K and 0.5 MPa undergoes isothermal reversible expansion until the pressure of the gas is 0.2 MPa. Which of the following option is correct? (log 2 = 0.3010, log 5 = 0.6989) (a) w = −9.1 kJ, ΔU = 0, ΔH = 0, q = 9.1 kJ (b) w = 9.1 J, ΔU = 9.1 J, ΔH = 0, q = 0 (c) w = +4.1 kJ, ΔU = 0, ΔH = 0, q = −4.1 kJ (d) w = −3.9 kJ, ΔU = 0, ΔH = 0, q = 3.9 kJ

20. C6XVCAT6Q35 16 g oxygen gas expands at STP (against a constant external pressure of 1 atm) to occupy double of its original volume. The work done during the process is : (a) 260 cal (b) 180 cal (c) 130 cal (d) 272.84 cal

23. C3XVCAT4Q28 If strength of a solution is 98 g/L and equivalent mass is 49, find normality. (A) 0.5 N (B) 1 N (C) 2 N (D) 3 N

24. C4XVRT3Q50 Among the following species, how many have their ionic size greater than S 2− ? The species are O 2− , P 3− , Cl − , S − , Ca 2+ , K + .

25. C5XVCAT5Q46 Consider the following order according to indicated property. (i) B–F bond length : BF_3 < BF_2OH (ii) Dipole moment : $CHCl_3$ < CH_3Cl (iii) Acidic strength : SO_3 < SeO_3 (iv) Number of d-orbitals participating in hybridisation : BrF_5 < PF_5. Find the total number of correct orders.

Mathematics

PART A · Q1–20 · SINGLE CORRECT

1. M1L2VSuggQ3 Find the measure of each interior angle of a regular octagon. (a) 120° (b) 135° (c) 140° (d) 150°

2. M2XVRT4Q60 Let X = {x : x = n^3 + 2n + 1, n ∈ N} and Y = {x : x = $3n^2$ + 7, n ∈ N}, then (a) X ∩ Y is a subset of {x : x = 3n + 5, n ∈ N} (b) X ∩ Y ⊆ {x : x = n^2 + n + 1, n ∈ N} (c) 34 ∈ X ∩ Y (d) none of these

3. M3SBVPYQQ7 A tower PQ stands on a horizontal ground with base Q on the ground. The point R divides the tower in two parts such that QR = 15 m. If from a point A on the ground the angle of elevation of R is 60° and the part PR of the tower subtends an angle of 15° at A, then the height of the tower is (a) 5(2√3 + 3) m (b) 5(√3 + 3) m (c) 10(√3 + 1) m (d) 10(2√3 + 1) m

4. M4L3V5Q5 The range of f(x) = [1 + sin x] + [2 + sin(x/2)] + [3 + sin(x/3)] + ... + [n + sin(x/n)], ∀ x ∈ [0, π], where [·] denotes the greatest integer function, is (a) {(n^2 + n − 2)/2, n(n+1)/2} (b) {n(n+1)/2} (c) {(n^2 + n − 2)/2, n(n+1)/2, (n^2 + n + 2)/2} (d) {n(n+1)/2, (n^2 + n + 2)/2}

5. M5CL2VSuggQ2 Using the half-angle identity, sin 22.5° equals: (a) √((1 − cos45°)/2) (b) √((1 + cos45°)/2) (c) sin45°/cos45° (d) √((1 − sin45°)/2)

6. M6L2VSuggQ2 Solve the system: x ≥ 3 and x < 7. (a) x ∈ (3, 7] (b) x ∈ [3, 7) (c) x ∈ (3, 7) (d) x ∈ [3, 7]

7. M7L2VSuggQ3 Using logarithmic rules, simplify: log3 27 + log3 9 − log3 81. (a) 1 (b) 2 (c) 3 (d) 0

8. M8AL3VSuggQ3 If a = 2, d = 2, verify t_4 = average of t_2 and t_6: (a) 8 (b) 10 (c) 12 (d) 6

9. M8AXVRT4Q59 If a_1, a_2, a_3, ... are in A.P. and a_4 + a_8 + a_{12} + a_{16} = 224, then sum of first nineteen terms is (a) 1540 (b) 1064 (c) 3125 (d) 1980

10. M8BL2V5Q6 The value of 4^(1/4) · 4^(1/8) · 4^(1/16) · 4^(1/32) .....∞ is (a) 2 (b) 3/2 (c) 1 (d) 1/2

11. M8BXVCAT5Q62 If a, b, c, d and p are distinct real numbers such that (a^2 + b^2 + c^2)p^2 − 2p(ab + bc + cd) + (b^2 + c^2 + d^2) ≤ 0, then a, b, c, d satisfy: (a) AP (b) GP (c) HP (d) ab = cd

12. M8CL1V6Q10 If a, b, c (all distinct positive) are in H.P., then the equation x^2 − kx + 2b^(101) − a^(101) − c^(101) = 0 (k ∈ R) has: (a) both roots positive (b) both roots negative (c) one positive and one negative root (d) both roots imaginary

13. M8DL2VSuggQ3 Find sum of 1 st two terms of AGP with a = 1, d = 1, r = 2 (a) 5 (b) 6 (c) 4 (d) 3

14. M8DL4V6Q5 If three positive real numbers a, b, c are in A.P. such that abc = 4, then the minimum possible value of b is (a) 2^(3/2) (b) 2^(2/3) (c) 2^(1/3) (d) 2^(5/2)

15. M8DSBVPYQQ4 Consider two G.Ps. 2, 2^2, 2^3, ... and 4, 4^2, 4^3, ... of 60 and n terms respectively. If the geometric mean of all the 60 + n terms is 2^(225/8), then Σ (k = 1 to n) k(n − k) is equal to: (a) 560 (b) 1540 (c) 1330 (d) 2600

16. M9L1V1Q5 Which of the following numbers is not a real number? (a) 2 (b) -1 (c) √2 (d) i

17. M9L2V5Q5 Find x + y if real numbers x & y satisfy the following equation: (x + 2y) + (2x − 3y)i + 4i = 5 (a) 1 (b) 2 (c) 3 (d) 4

18. M9L3V2Q4 If z is a complex number then $z + \bar{z}$ = (a) Re(z) (b) 2Re(z) (c) Im(z) (d) 2Im(z)

19. M9L4V6Q8 If ω = α + iβ, where β ≠ 0, and z ≠ 1, satisfies the condition that $((\omega - \omega\bar{z})/(1 - z))$ is purely real, then the set of values of z is (a) {z : |z| = 1} (b) {z : z = $\bar{z}$} (c) {z : z ≠ 1} (d) {z : |z| = 1, z ≠ 1}

20. M9L5VSuggQ7 If z = (√3+i)/(√3-i), then the fundamental amplitude of z is : (a) -π/3 (b) π/3 (c) π/6 (d) None

PART B · Q21–25 · NUMERICAL VALUE

21. M4XVRT3Q72 If f(x) = 3[x + 1] + 2[x + 2] + [x − 5], then the value of f(2.999) is (where [.] denotes the greatest integer function)

22. M6SBVPYQQ4 Consider the function f : R → R defined by f(x) = 2x/√(1 + 9x^2). If the composition of f, (f∘f∘f∘...∘f)(x) [f taken 10 times] = 2^(10) x/√(1 + 9αx^2), then the value of √(3α + 1) is equal to _____

23. M7L1V4Q2 If (√(√2))^(1/2) = 2^λ, then the value of 64λ is:

24. M7XVCAT4Q74 Let P = 5 / ($1/\log_2 x + 1/\log_3 x + 1/\log_4 x + 1/\log_5 x$) and (120)^P = 32, then the value of x be:

25. M8BXVCAT5Q71 If Σ_(k=1)^10 (2^k + 3^(k−1)) = (1/2)(2^(λ+2) + 3^λ − μ), find μ.